

JKBOSE 2026

CLASS 11TH

#XUV ★❤️

CHEMICAL BONDING AND MOLECULAR STRUCTURE

LECTURE-5

CHEMISTRY

VISHAL SIR

Bhaiichara
ON
TOP



JK Bole Wallah

Topics To Covered

1

Valence Shell Electron Pair Repulsion (VSEPR) Theory

REVISION.

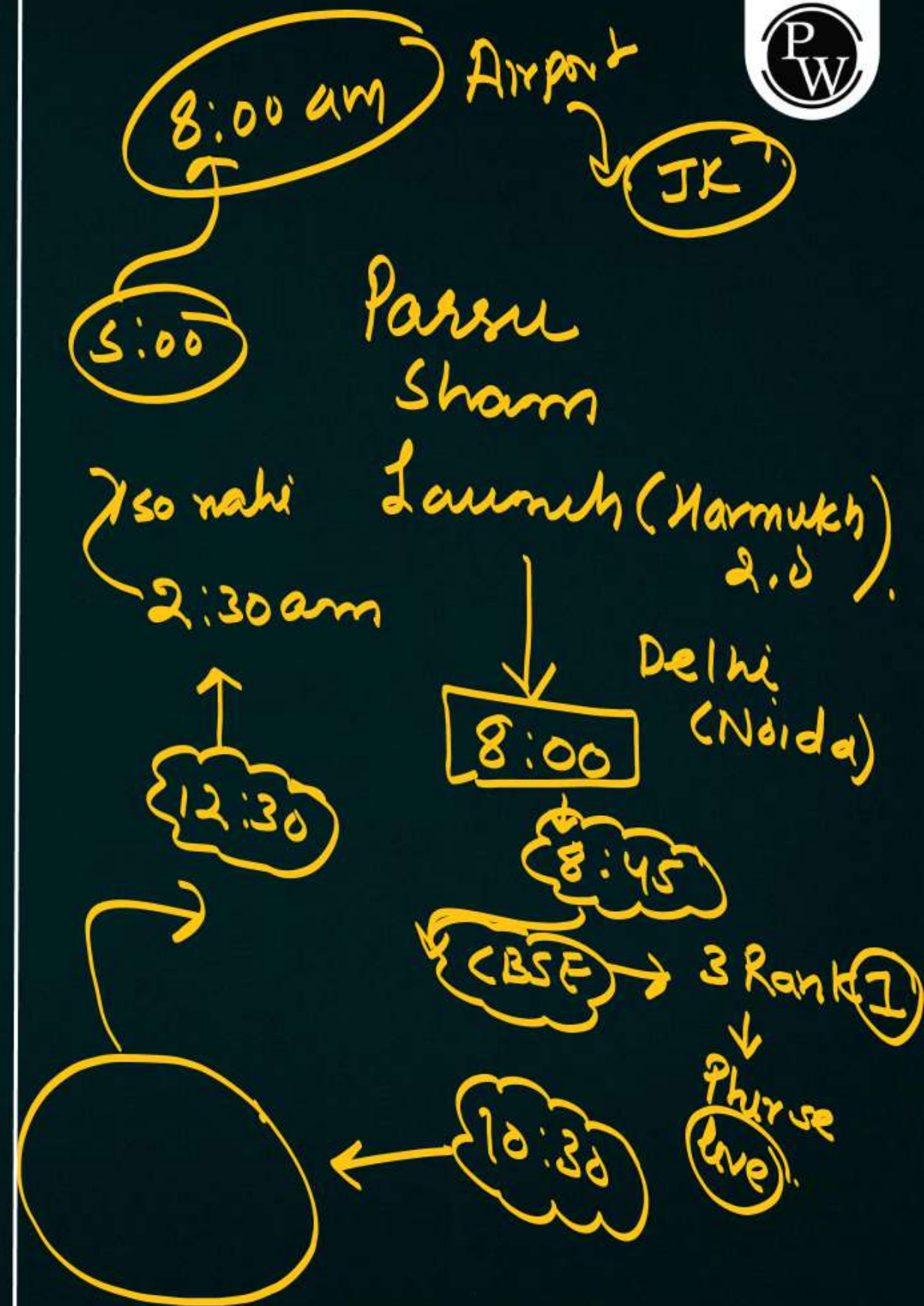
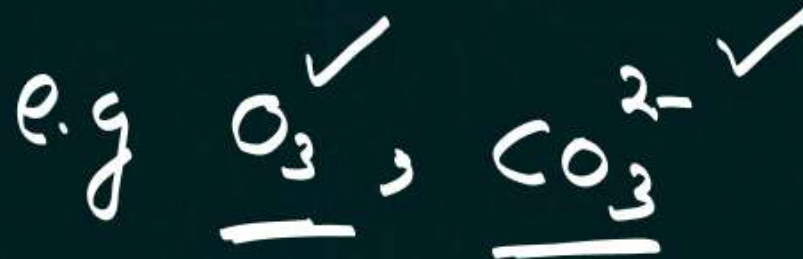
BOND Parameters $\rightarrow$

- a) Bond length.
- b) Bond enthalpy.
- c) Bond angle
- d) Bond orders.

Resonance

Resonance energy

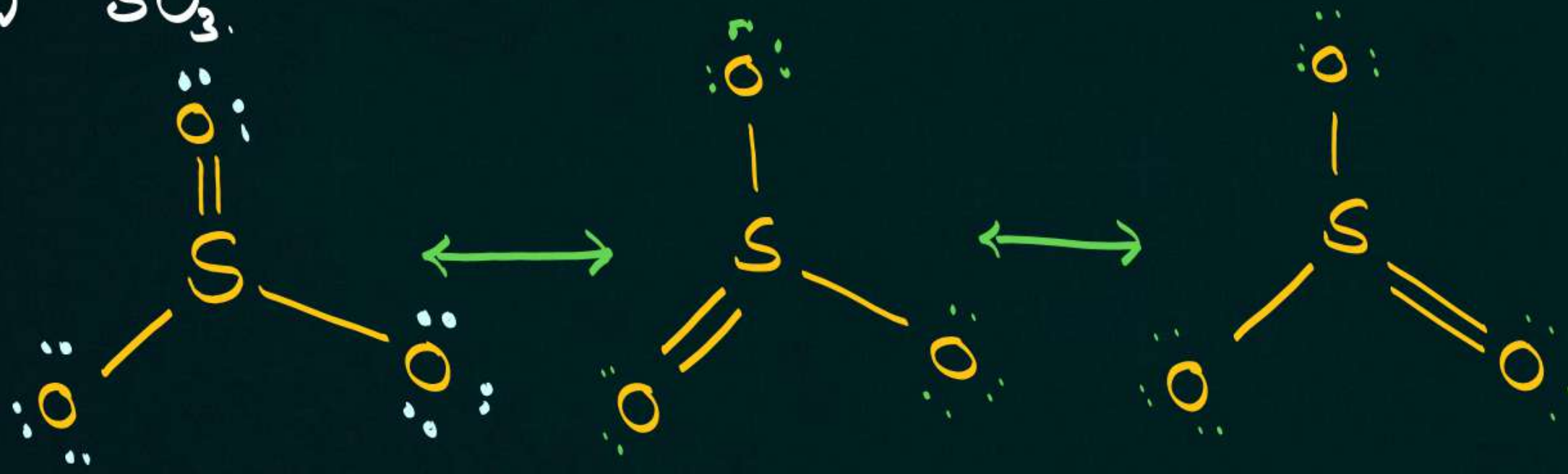
Rules $\rightarrow$



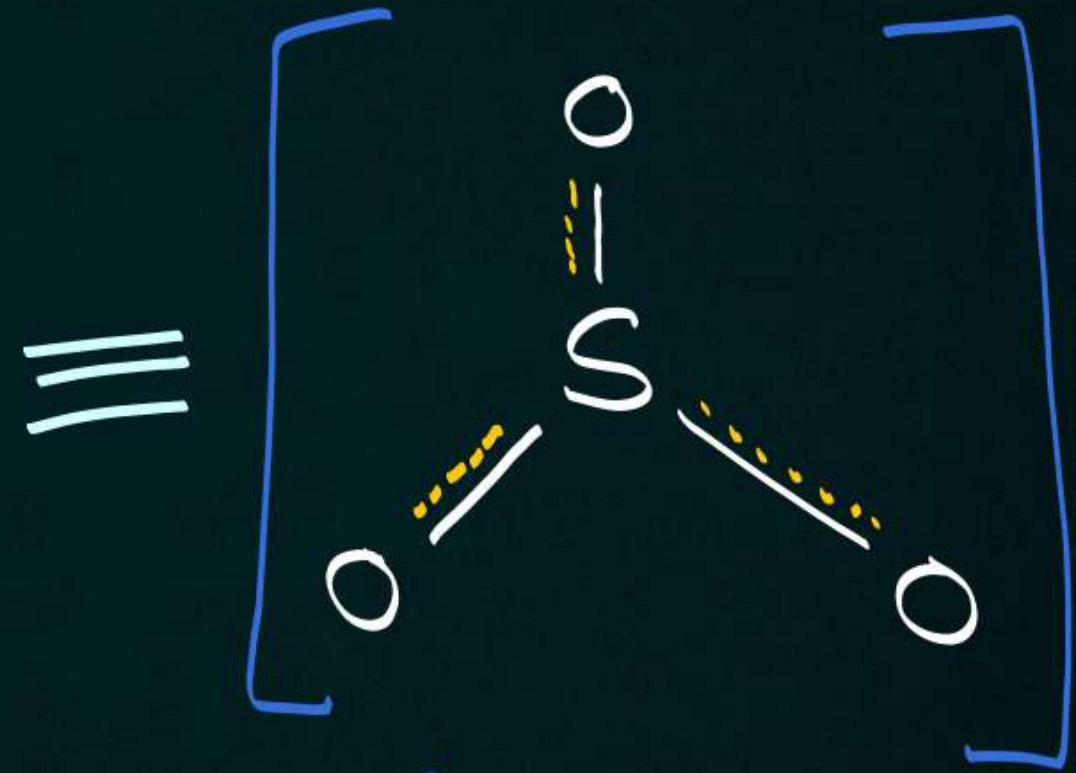
Q. Draw the Resonating str. of SO_3 , NO_2 & NO_3^- .

Ans

i) SO_3

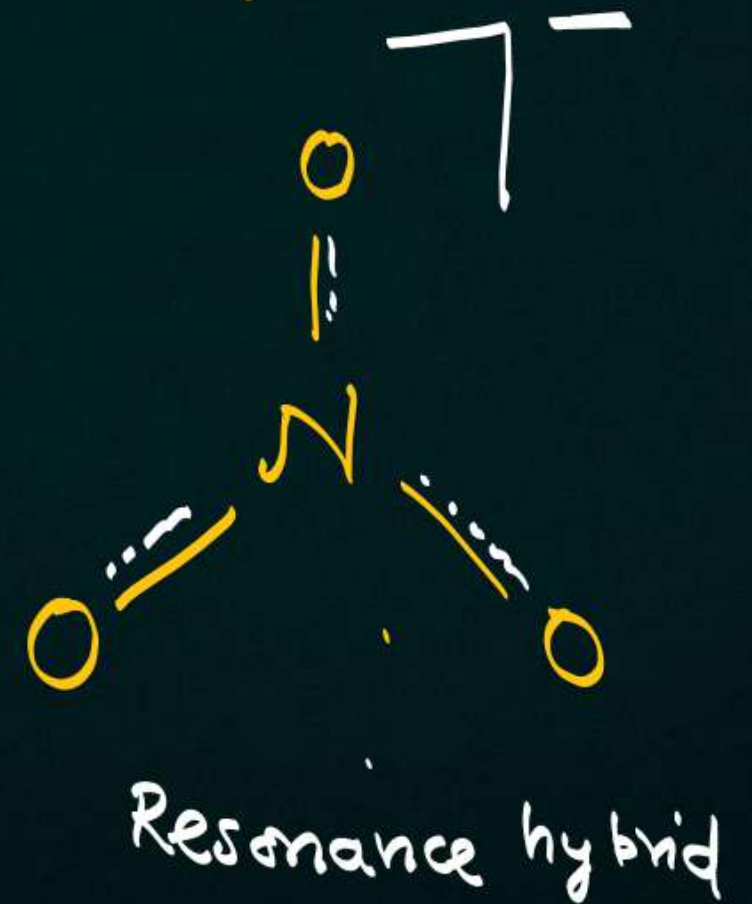
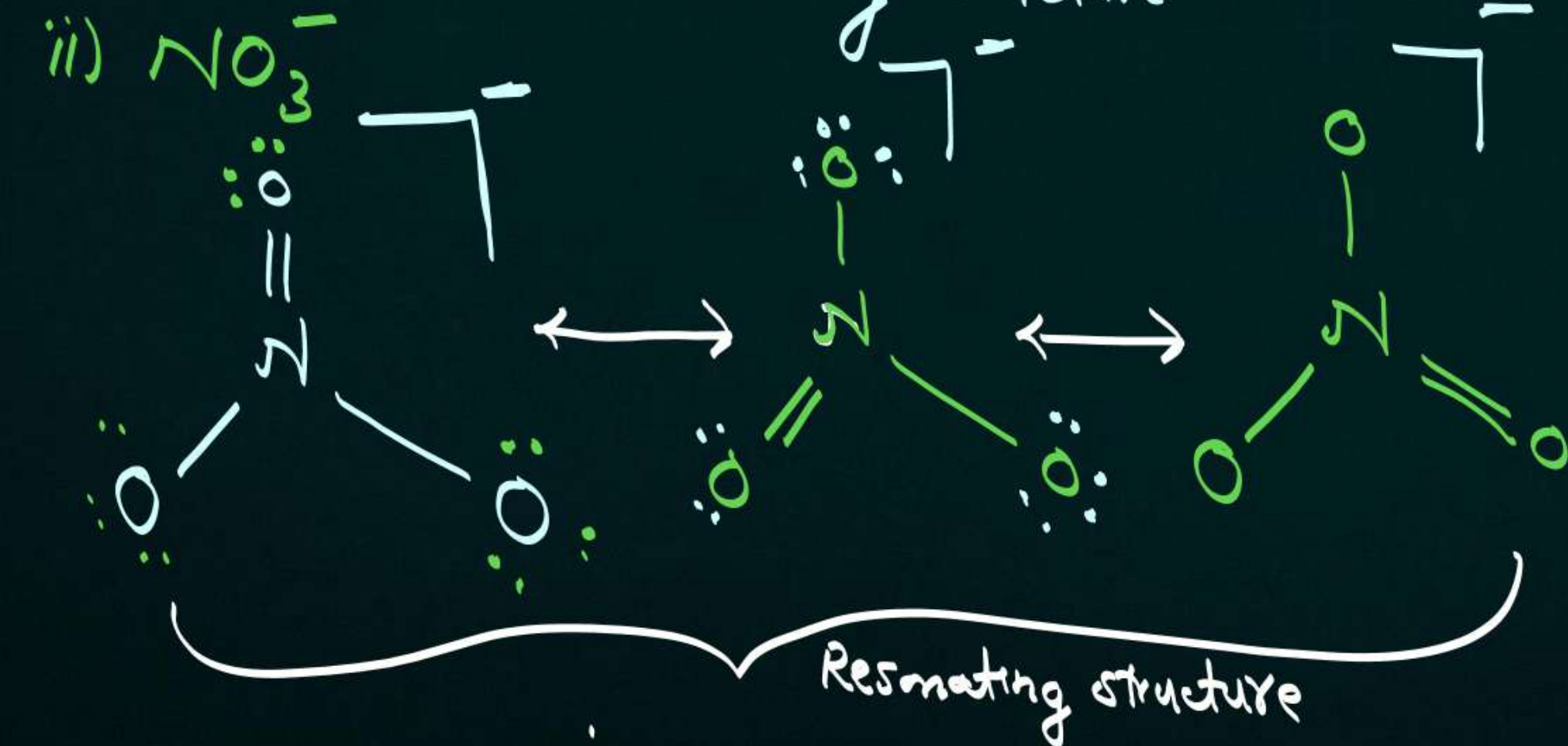
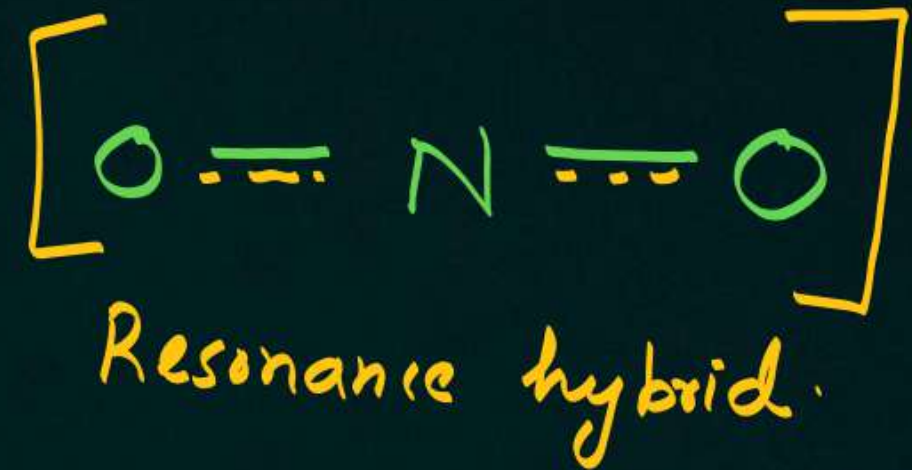
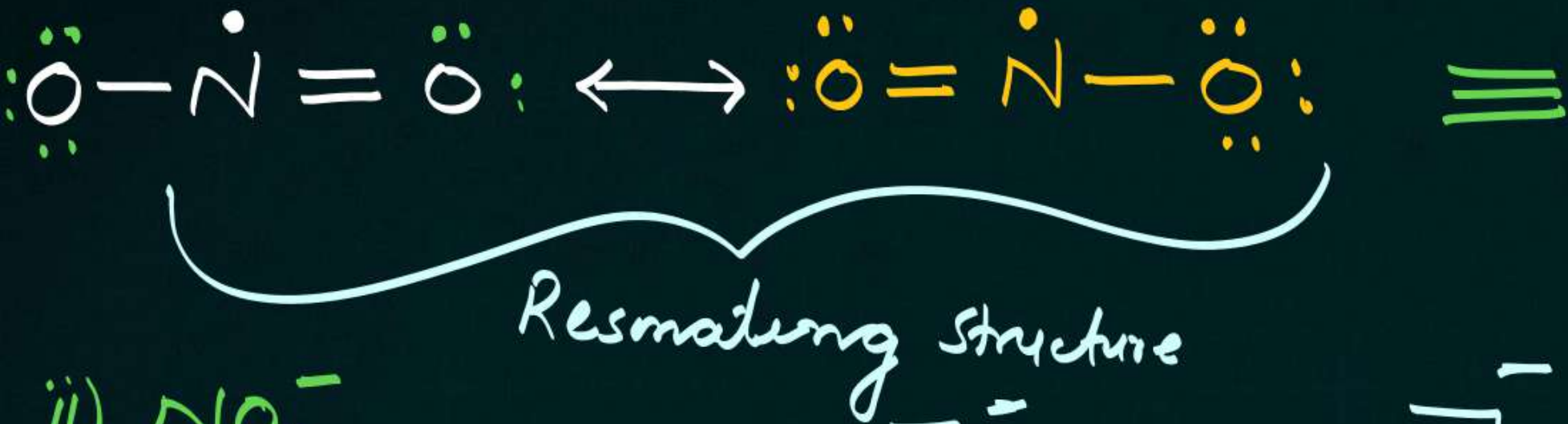


Resonating str.



Resonance hybrid.

ii) NO_2



Difference between physical properties of Ionic & Covalent Compounds.

IONIC COMPOUNDS.

1) Physical State → Due to strong force of attraction present between ionic compounds, they are solid in nature.

2) Melting & Boiling Point → Due to strong inter-molecular force of attraction, the melting & boiling point of ionic compounds is very high.

3) Electrical conductivity → In solid state, they are bad conductors of electricity but in molten state & in aqueous medium behave as good conductor of electricity due to the presence of ions.

COVALENT COMPOUNDS.

1) Physical State → Force of attractions are of an moderate range between the molecules. Covalent compounds can be solid (I_2), liquid (Br_2) & Gases (Cl_2).

2) Melting & Boiling Point → Due to moderate force of attraction in covalent compounds, the melting & boiling point are comparatively low.

3) Electrical conductivity → Bad conductors of electricity.

4) Solubility $\Rightarrow$ Easily soluble in water due to polar nature of ionic compounds.

4) Solubility $\Rightarrow$ Polar covalent compounds easily dissolve in water & non-polar covalent compounds are insoluble in water.

Valence Shell Electron Pair Repulsion (VSEPR) Theory



- **Idea given by:** Sidgwick and Powell in 1940.
- **Improved by:** Nyholm and Gillespie in 1957.
- **Basic concept:** The shape of a molecule depends on how electron pairs (bonded or lone pairs) arrange themselves around the central atom

Main points of VSEPR theory:

1. The shape of a molecule depends on the number of electron pairs (bond pairs + lone pairs) around the central atom.
2. Since electron clouds are negatively charged, they repel each other.
3. To reduce repulsion, electron pairs try to stay as far apart as possible.
4. We imagine the valence shell as a sphere, and the electron pairs spread out on its surface at maximum distance.
5. A double or triple bond is treated like **one single pair** (called a super pair).

★ 6.) Order of repulsion of e-pairs
 $lp-lp > lp-bp > bp-bp$

i) Shape of molecule $\rightarrow$ $\bar{e}$ -pairs arrangement.

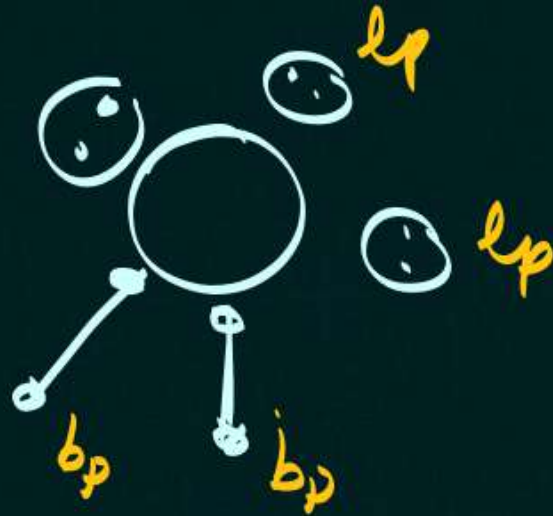
ii) $\bar{e}$ pair repel.

iii) $\bar{e}$ -pair maximum distance

iv) Central atom
Valence shell $\rightarrow$ sphere

v) $=$ or $\equiv$ equal single bond

* vi) $lp-lp > lp-bp > bp-bp$



Predicting the Shapes of Molecules



1) Total electron pairs (TEP) = $\frac{1}{2}(V + M - C + A)$.

V = Valence e of central atom.

M = No. of Monovalent atom (except O & Nitrogen family)

C = Total charge on cation

A = Total " " Anion.

NOTE

Geometry ↓ lone pair + bond pair	Shape ↓ only bond pairs.
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O
S
Se
Te
Po
Lv
 N
P
As
Sb
Bi
Mc

How to decide geometry $\rightarrow$

TEP

Geometry.

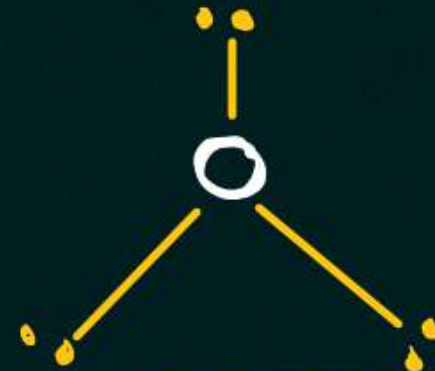
2.

Linear.



3

Trigonal planar.

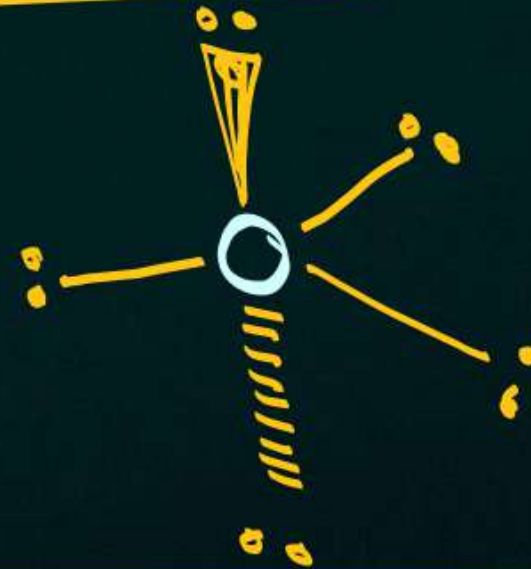


Tetrahedral.



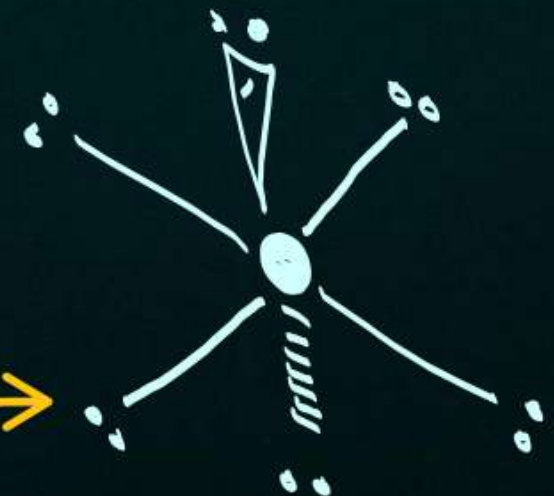
4

Trigonal bipyramidal



5

Octahedral.



6

Predicting the Shapes of Molecules

A)

$$\text{TEP} = 2 \quad (\text{lp} + \text{bp})$$

Case: 0 lp + 2 bp

Geometry $\Rightarrow$ linear.

Shape $\Rightarrow$ linear.

e.g. BeF_2 , BeCl_2 .



B)

$$\text{TEP} = 3$$

i.e. $\text{lp} + \text{bp} = 3$.

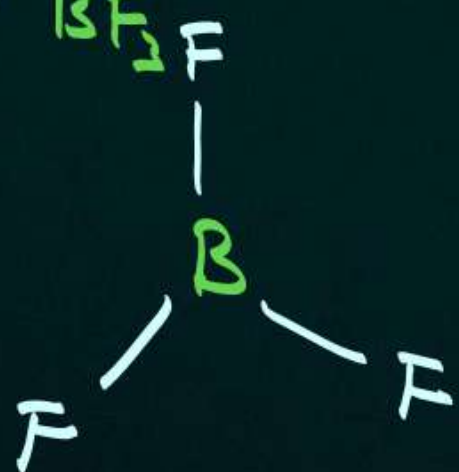
Geometry $\Rightarrow$ Trigonal planar.



0 lp + 3 bp

Shape: Trigonal planar

e.g. BF_3



1 lp + 2 bp.

Shape $\Rightarrow$ Bent.

e.g. O_3 .





$$\begin{aligned} \text{TEP} &= \frac{1}{2}(V + M - C + A) \\ &= \frac{1}{2}(2 + 2) = \frac{4}{2} = \textcircled{2} \end{aligned}$$

TEP = 2.
Geometry $\rightarrow$ Linear.

Bond pair = no. of side atom
= 2.

$$\begin{aligned} \text{lone pair} &= \text{TEP} - \text{BP} \\ &= 2 - 2 = 0 \end{aligned}$$

$$\underline{\text{alp}} + 2\text{bp}$$

Shape: Linear $\text{F}-\text{Be}-\text{F}$



$$\begin{aligned} \text{TEP} &= \frac{1}{2}(V + M - C + A) \\ &= \frac{1}{2}(3 + 3) = \frac{6}{2} = 3 \end{aligned}$$

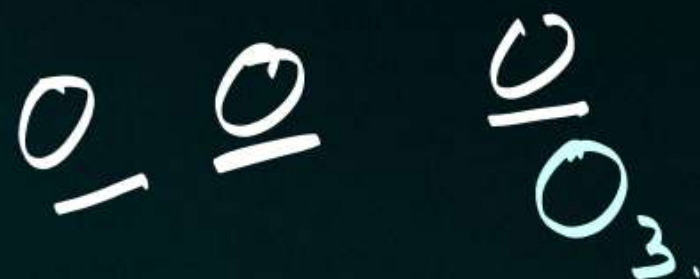
TEP = 3, Geometry $\rightarrow$ Trigonal planar

Shape $\rightarrow$

$$\text{bp} = 3$$

$$\text{lp} = \text{TEP} - \text{bp} = 3 - 3 = 0$$

Shape: Trigonal planar



$$\begin{aligned} \text{TEP} &= \frac{1}{2}(V + M - C + A) \\ &= \frac{1}{2}(6 + 0) = \frac{6}{2} = \textcircled{3} \end{aligned}$$

Geometry $\rightarrow$ Trigonal planar.

Shape $\rightarrow$

$$\text{bp} = 2$$

$$\text{lp} = \text{TEP} - \text{bp} = 3 - 2 = 1$$

Shape: Bent.



H₂O

$$TEP = \frac{1}{2}(v + m - c + a).$$

$$= \frac{1}{2}(6 + 2) = \frac{8}{2} = \textcircled{4}.$$

QUESTION



Discuss the shape of the following molecules using VSEPR model



TLP only

Y.W

REVISE KAR K HI SONA

Thank
You

Hybri